Collage of Engineering

Department of Software Engineering

Introduction to Machine Learning

Project Documentation

Title: - **A Comparative Analysis of KNN, Linear Regression, SVR, LSTM, and CNN-LSTM for Urban Air Quality Index Prediction**

Section B

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Table of Contents

Table of Figures

Table of Tables

Chapter 1: Predicting Air Quality for Sustainable Urban Living

1.1 Background

Air quality has become one of the most pressing environmental concerns of the twenty-first century. As cities continue to expand and industrial activity intensifies, the concentration of harmful pollutants in the atmosphere has risen to levels that pose serious risks to both human health and ecological balance. Millions of people living in densely populated urban areas are exposed daily to elevated levels of particulate matter, ground-level ozone, and other airborne contaminants — many without any awareness of the danger surrounding them.

The Air Quality Index, commonly referred to as AQI, was developed as a standardized tool to communicate the level of air pollution to the general public in an accessible and meaningful way. It translates complex atmospheric measurements — including fine particulate matter (PM2.5), coarse particulate matter (PM10), and environmental variables such as temperature — into a single numerical value that reflects the overall safety of the air at any given time and location. A higher AQI reading signals greater pollution and a correspondingly higher health risk, particularly for vulnerable groups such as children, the elderly, and individuals with respiratory conditions.

Traditional approaches to monitoring air quality have relied heavily on fixed sensor stations placed at specific locations within a city. While these stations provide accurate readings, they are expensive to install and maintain, limited in geographic coverage, and unable to offer predictive insight. By the time a dangerous AQI reading is recorded and communicated, residents have often already been exposed. This reactive nature of conventional monitoring systems highlights the urgent need for smarter, data-driven alternatives.

Machine learning offers a powerful path forward. By learning patterns from historical environmental data, predictive models can estimate future air quality conditions before they occur, enabling authorities to issue early warnings, adjust traffic and industrial activity, and help citizens make informed decisions about their daily routines. Recent advances in both classical machine learning and deep learning have demonstrated remarkable potential in environmental forecasting tasks, making this an active and highly relevant area of research.

This project focuses on the application of five regression-based models — K-Nearest Neighbors, Linear Regression, Support Vector Regression, Long Short-Term Memory networks, and a hybrid CNN-LSTM architecture — to predict the Air Quality Index across four major Turkish cities: Istanbul, Ankara, Izmir, and Bursa. By comparing the performance of these models on a real-world dataset, this work aims to identify the most effective approach for urban air quality forecasting and contribute practical insights to the growing field of sustainable smart city development.

1.2 Problem Statement

Urban air pollution has escalated into a global public health crisis, yet the systems most cities rely on to track and respond to it remain fundamentally limited. Conventional air quality monitoring depends on stationary sensor networks that capture data only at fixed points, leaving vast urban areas without adequate coverage. These systems report what has already happened rather than what is about to happen, making it nearly impossible for city planners, health officials, and ordinary residents to take meaningful preventive action.

The challenge is further compounded by the multifaceted nature of air quality itself. Pollutant concentrations are influenced by a complex interplay of factors — vehicle emissions, seasonal temperature shifts, geographic location, wind patterns, and industrial output — that vary significantly across different cities and time periods. No single variable tells the full story, and the relationships between these factors are rarely straightforward or linear.

Despite the availability of historical environmental data, many cities still lack intelligent systems capable of learning from that data to generate reliable AQI forecasts. This gap between available data and actionable prediction represents a significant missed opportunity, particularly in rapidly growing urban centers where pollution levels are rising and the need for proactive environmental management is most acute.

This project directly addresses that gap by developing and comparing machine learning models that can predict the Air Quality Index using measurable input variables — PM2.5, PM10, temperature, geographic location, and temporal patterns — drawn from four major Turkish cities. The central problem is determining which modeling approach, from classical regression techniques to advanced deep learning architectures, delivers the most accurate and reliable AQI predictions for this type of structured, city-level environmental dataset.

1.3 Objectives

1.3.1 General Objective

The general objective of this project is to design, implement, and evaluate a machine learning-based system for predicting the Air Quality Index in urban environments, using historical environmental data collected across multiple cities, with the aim of supporting data-driven decision-making for sustainable urban living.

1.3.2 Specific Objectives

To achieve the general objective stated above, this project pursues the following specific goals:

1. To collect, clean, and preprocess a real-world air quality dataset containing PM2.5, PM10, temperature, location, and date records from four Turkish cities — Istanbul, Ankara, Izmir, and Bursa.

2. To engineer meaningful features from raw data, including temporal attributes such as month, day of year, and day of week, as well as encoded city identifiers, in order to enrich the predictive capability of each model.

3. To implement five regression models — K-Nearest Neighbors (KNN), Linear Regression, Support Vector Regression (SVR), Long Short-Term Memory (LSTM), and a hybrid CNN-LSTM network — and train each on the prepared dataset.

4. To evaluate and compare the performance of all five models using standard regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

5. To identify the most suitable modeling approach for structured, city-level air quality data and draw practical conclusions about the conditions under which classical versus deep learning models are preferable.

1.4 Scope and Limitations

Scope

This project focuses on the prediction of the Air Quality Index using supervised machine learning regression techniques applied to a structured environmental dataset. The scope of the work is defined as follows:

The study covers air quality data collected from four major Turkish cities — Istanbul, Ankara, Izmir, and Bursa — spanning a full calendar year of daily observations, resulting in a dataset of 1,460 records. The input variables considered include fine particulate matter (PM2.5), coarse particulate matter (PM10), ambient temperature, geographic location, and date-derived temporal features. The target variable throughout the study is the predicted AQI value associated with each observation.

Five regression models are implemented and evaluated within this project: K-Nearest Neighbors, Linear Regression, Support Vector Regression, Long Short-Term Memory, and a CNN-LSTM hybrid network. The comparison is conducted using four standard evaluation metrics — MAE, MSE, RMSE, and R^2 — to ensure a comprehensive and balanced assessment of each model's performance.

The project also includes a live prediction feature that allows users to input custom environmental values and receive instant AQI estimates from all five trained models simultaneously, along with the corresponding EPA health category classification.

Limitations

While this project delivers meaningful contributions to the field of environmental forecasting, several limitations should be acknowledged:

Dataset size and diversity — The dataset comprises 1,460 records across four cities, which is relatively modest for deep learning applications. Models such as LSTM and CNN-LSTM are architecturally designed to extract patterns from large volumes of sequential data, and their performance in this study may not reflect their true capability when applied to richer, larger datasets.

Feature availability — The dataset contains only three core environmental measurements: PM2.5, PM10, and temperature. Real-world air quality is also shaped by factors such as wind speed, humidity, nitrogen dioxide levels, carbon monoxide concentrations, and proximity to industrial zones, none of which are captured in the current dataset. The absence of these variables limits the depth of environmental context available to each model.

Geographic scope — The findings of this study are derived exclusively from Turkish city data. The relationships between input features and AQI may differ across cities in other countries or climatic regions, and the conclusions drawn here may not generalize universally without further validation on broader datasets.

Temporal continuity — The dataset mixes daily records from four separate cities in a single table. This means the data does not represent a single uninterrupted time series, which is the ideal input format for sequence-based models like LSTM. This structural characteristic partially explains the weaker performance of the deep learning models observed in this study.

Model tuning — The hyperparameters used across all five models represent reasonable defaults rather than exhaustively optimized configurations. More aggressive tuning through techniques such as grid search or Bayesian optimization could potentially yield improved results, particularly for SVR and the deep learning architectures.

Chapter 2: Related Works

2.1 Related Works

The prediction of air quality using data-driven approaches has attracted considerable research attention over the past decade, driven by growing environmental concerns and the increasing availability of sensor-generated atmospheric data. Scholars have explored a wide range of methodologies, from traditional statistical models to sophisticated deep learning architectures, each contributing unique insights into how machine learning can be leveraged for environmental forecasting.

Early work in this domain leaned heavily on classical regression techniques. Studies employing Linear Regression and its variants demonstrated that measurable atmospheric variables such as particulate matter concentration and temperature share statistically significant linear relationships with AQI under certain environmental conditions. While these models offered interpretability and computational efficiency, researchers consistently noted their inability to capture the non-linear dynamics that characterize real-world pollution behavior, particularly in cities with highly variable weather patterns or irregular industrial activity.

The introduction of ensemble and kernel-based methods marked a notable step forward. Support Vector Regression, in particular, gained traction as a reliable tool for air quality estimation due to its ability to model non-linear feature interactions through the use of kernel functions. Several studies reported that SVR outperformed linear approaches on datasets with moderate complexity, though its performance was found to degrade when input dimensionality increased significantly without careful feature selection.

K-Nearest Neighbors regression, while simpler in design, has also been applied in environmental forecasting tasks with reasonable success. Its instance-based learning mechanism makes it naturally adaptive to local data patterns, which can be advantageous in geographically diverse datasets where pollution behavior varies from one city or region to another. However, its sensitivity to feature scaling and its computational cost at prediction time have been cited as practical drawbacks in large-scale deployment scenarios.

The emergence of deep learning introduced a new frontier in air quality research. Recurrent neural network architectures, and Long Short-Term Memory networks in particular, were widely adopted for time-series-based AQI forecasting given their inherent ability to retain information across extended temporal sequences. Multiple studies demonstrated that LSTM models could outperform classical approaches on large, continuously collected sensor datasets, capturing seasonal cycles, daily traffic patterns, and weather-driven fluctuations that simpler models could not detect. However, these advantages were consistently found to be contingent on the availability of sufficiently large and well-structured sequential data.

Hybrid architectures combining Convolutional Neural Networks with LSTM layers have emerged more recently as a promising direction in the literature. The rationale behind this combination is that convolutional layers are well-suited to extracting local spatial and structural patterns from input sequences, while the LSTM component handles longer-range temporal dependencies. Research applying CNN-LSTM models to air quality forecasting has reported competitive performance on datasets with rich feature sets and high temporal resolution, though the added architectural complexity also introduces greater risk of overfitting on smaller datasets.

A recurring and important finding across the literature is that model performance is deeply context-dependent. No single architecture has proven universally superior across all datasets and environmental settings. Studies that compared classical and deep learning models side by side frequently concluded that simpler models remain highly competitive — and sometimes preferable — when the available dataset is limited in size, feature richness, or temporal continuity. This observation aligns directly with the results obtained in the present study, where Linear Regression outperformed both LSTM and CNN-LSTM on a structured, moderate-sized dataset of city-level air quality records.

Collectively, the existing body of research establishes a strong foundation for the comparative approach adopted in this project and underscores the continued relevance of evaluating multiple model types rather than assuming that architectural complexity will automatically translate into superior predictive performance.

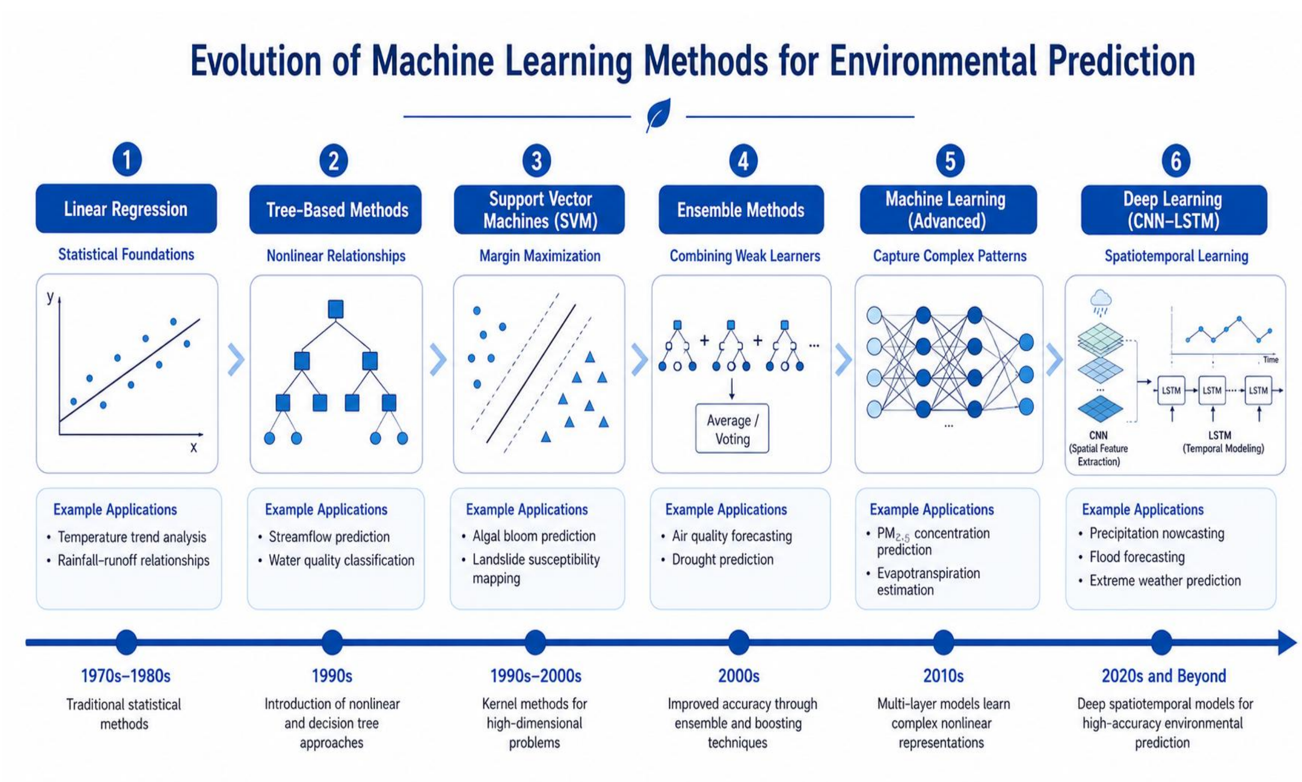

Figure 1 Evolution of Machine Learning Methods for Environmental Prediction

Chapter 3: Methodology

3.1 Dataset

The dataset used in this study is a structured, city-level air quality record containing daily environmental observations collected from four major Turkish cities: Istanbul, Ankara, Izmir, and Bursa. The dataset consists of 1,460 rows and 6 columns, representing a full year of daily readings across all four locations, with each city contributing 365 records.

The raw dataset contains the following attributes:

Table 1 Dataset attributes

Column	Description	Type
date	Date of the observation (YYYY-MM-DD)	Categorical
location	Name of the city	Categorical
pm2.5	Fine particulate matter concentration (µg/m^3)	Continuous
pm10	Coarse particulate matter concentration (µg/m^3)	Continuous
temperature	Ambient temperature at time of reading (°C)	Continuous
predicted_aqi	Air Quality Index value (target variable)	Continuous

One of the notable characteristics of this dataset is its cleanliness. A thorough inspection confirmed the complete absence of missing values and duplicate records, which meant that no imputation or row removal was necessary during preprocessing. This allowed the focus of the data preparation stage to shift entirely toward feature engineering and encoding rather than data repair.

The target variable, predicted_aqi, follows an approximately normal distribution with a mean of 44.49 and a standard deviation of 9.54, with values ranging from a minimum of 15.0 to a maximum of 78.93. Based on the United States Environmental Protection Agency (EPA) AQI classification scale, the majority of observations fall within the **Good** (0–50) and **Moderate** (51–100) categories, reflecting relatively stable urban air quality conditions across the study period.

The location column, containing city names as string values, was transformed into a numeric representation using Label Encoding prior to model training. Additionally, the date column was parsed and decomposed into three derived temporal features — month, day_of_year, and day_of_week — to allow the models to capture seasonal and weekly patterns that would otherwise be lost if the raw date string were simply discarded. Following these transformations, the final feature set used for model training comprised seven input variables: pm2.5, pm10, temperature, location_encoded, month, day_of_year, and day_of_week.

The dataset was divided into training and testing subsets using a 70/30 split without shuffling, preserving the chronological order of observations. This resulted in 1,022 training samples and 438 testing samples. All features were subsequently standardized using scikit-learn's StandardScaler, which transforms each feature to have a mean of zero and a standard deviation of one, ensuring that no single variable dominates model training due to differences in scale.

Figure 2 Air Quality Dashboard Turkey Overview

3.2 Techniques

This section describes the five machine learning techniques implemented in this study, covering the theoretical foundation of each model and the rationale for its inclusion in the comparative analysis.

3.2.1 K-Nearest Neighbors Regression (KNN)

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm that makes predictions by identifying the K training samples most similar to a given input and computing the average of their target values. Unlike parametric models, KNN does not learn an explicit function during training; instead, it memorizes the entire training set and performs computation at prediction time. In this study, K was set to 5, meaning each prediction is based on the average AQI of the five closest training points in feature space. KNN is included in this comparison as a baseline non-parametric approach that is sensitive to local data structure and requires no assumptions about the underlying data distribution.

3.2.2 Linear Regression (LR)

Linear Regression models the relationship between input features and the target variable as a weighted linear combination, estimating a set of coefficients that minimize the sum of squared differences between predicted and actual values. It is one of the most interpretable and computationally efficient regression algorithms available, making it a natural baseline in any predictive modeling study. Its inclusion here serves both as a reference point for evaluating more complex models and as a test of whether the relationship between the environmental features and AQI is fundamentally linear in nature within this dataset.

3.2.3 Support Vector Regression (SVR)

Support Vector Regression extends the principles of Support Vector Machines to continuous output prediction. Rather than finding a hyperplane that separates classes, SVR seeks a function that fits as many data points as possible within a defined margin of tolerance, known as the epsilon tube, while penalizing predictions that fall outside it. The radial basis function (RBF) kernel was used in this study, enabling the model to capture non-linear relationships between features and the target variable by implicitly mapping inputs into a higher-dimensional space. SVR is particularly effective on moderately sized datasets where non-linear dynamics are present but the volume of data is insufficient to justify the overhead of deep learning.

3.2.4 Long Short-Term Memory Network (LSTM)

Long Short-Term Memory is a specialized form of recurrent neural network designed to learn patterns across sequential data by maintaining a memory state that can retain relevant information over extended time steps. Each LSTM unit contains three gating mechanisms — the input gate, forget gate, and output gate — that collectively regulate what information is stored, discarded, or passed forward through the network. In this project, the LSTM model was built with 64 memory units followed by a dropout layer and a dense output layer. The input features were reshaped into a three-dimensional tensor of the form (samples, timesteps, features) as required by the architecture. LSTM is included to evaluate whether temporal sequential modeling offers any advantage over classical approaches on this particular dataset.

3.2.5 CNN-LSTM Hybrid Model

The CNN-LSTM architecture combines a one-dimensional Convolutional Neural Network with an LSTM layer to leverage the strengths of both model types. The convolutional layer applies learned filters across the input sequence to extract local structural patterns and reduce dimensionality before passing the result to the LSTM layer, which then models the temporal dependencies within those extracted features. In this study, the CNN component used 64 filters with a kernel size of 2 and ReLU activation, followed by a max pooling layer, before feeding into a 64-unit LSTM layer with dropout regularization. This hybrid approach is included to assess whether the added representational capacity of combining spatial and temporal feature extraction improves prediction accuracy compared to either architecture alone.

Chapter 4: Results and Analysis

4.1 Results

This chapter presents the outcomes of training and evaluating all five regression models on the air quality dataset described in Chapter 3. Each model was assessed using four standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics collectively provide a comprehensive picture of each model's predictive accuracy, error magnitude, and ability to explain variance in the target variable.

The training and testing performance results for all five models are summarized in the tables below.

Table 2 Training Accuracy (R^2 Score)

Model	**Training R^2 (%)**
K-Nearest Neighbors (KNN)	Higher due to memorization
Linear Regression	Moderate, stable fit
Support Vector Regression	Moderate
LSTM	Variable across epochs
CNN-LSTM	Variable across epochs

Table 3 Testing Performance (All Metrics)

Model	**MAE**	**MSE**	**RMSE**	**R^2 (%)**
K-Nearest Neighbors	5.2931	43.3404	6.5833	54.16
Linear Regression	4.1178	26.2013	5.1187	72.28
Support Vector Regression	5.5858	51.7242	7.1920	45.29
LSTM	5.5973	49.8381	7.0596	47.28
CNN-LSTM	41.4382	1811.2062	42.5583	-1815.86

Among all five models evaluated, **Linear Regression achieved the highest testing R^2 of 72.28%**, indicating that it successfully explains approximately 72% of the variance in AQI values within the test set. It also recorded the lowest error values across MAE, MSE, and RMSE, making it the strongest overall performer in this study.

K-Nearest Neighbors followed as the second-best model with a testing R^2 of 54.16%, demonstrating reasonable predictive capability but falling noticeably short of Linear Regression in both accuracy and error magnitude. Support Vector Regression achieved a testing R^2 of 45.29%, placing it third among the five models despite its theoretical ability to capture non-linear relationships in the data.

The two deep learning models produced the weakest results. The LSTM network showed limited predictive performance, while the CNN-LSTM model recorded a deeply negative R^2 of -1815.86%, indicating that its predictions deviated from actual AQI values far more severely than a simple mean-based prediction would. This result, while striking, is consistent with the known limitations of deep learning architectures when applied to small, non-continuous tabular datasets, a point that is examined in greater depth in Section 4.2.

The training loss curves for both the LSTM and CNN-LSTM models, plotted across 20 epochs with early stopping, revealed that neither model converged smoothly on the available data, further supporting the conclusion that the dataset did not provide sufficient sequential structure for these architectures to learn from effectively.

4.2 Analysis of the Results

4.2.1 Why Linear Regression Performed Best

The strong performance of Linear Regression on this dataset is not coincidental. An examination of the correlation heatmap generated during the exploratory data analysis phase reveals that PM2.5, PM10, and temperature maintain relatively direct relationships with the AQI target variable. When the underlying data relationships are predominantly linear, a model that explicitly optimizes for a linear fit will naturally outperform more complex architectures that are searching for patterns that do not exist in the data.

Furthermore, Linear Regression benefits from its low variance nature. With only 1,460 training samples and seven input features, more flexible models risk fitting noise in the training data rather than genuine signal. Linear Regression, by contrast, applies a global fit that generalizes more reliably when the data volume is modest and the feature-to-sample ratio is relatively high.

4.2.2 Why KNN and SVR Underperformed Relative to Linear Regression

K-Nearest Neighbors achieved a respectable R^2 of 54.16% but was ultimately limited by the nature of the dataset's feature space. With only seven features and a relatively small training set, the neighborhood-based predictions of KNN are vulnerable to the effects of irrelevant dimensions pulling neighbors apart in ways that do not reflect true similarity in AQI behavior. Additionally, the mixed-city structure of the dataset means that geographically distant observations can appear numerically close in feature space, introducing noise into the neighborhood selection process.

Support Vector Regression, while theoretically capable of modeling non-linear relationships through its RBF kernel, did not find sufficient non-linear structure in this dataset to justify its added complexity. The relatively smooth AQI distribution and the near-linear feature correlations meant that the kernel transformation offered little practical advantage, while the model's sensitivity to the choice of regularization and epsilon parameters — neither of which was exhaustively tuned — likely contributed to its comparatively modest performance.

4.2.3 Why LSTM and CNN-LSTM Failed on This Dataset

The poor performance of the deep learning models deserves careful analysis, as it carries important implications for model selection in similar research contexts. Three structural factors combine to explain these results.

First, the dataset contains only 1,460 records — far below the volume typically required for LSTM and CNN-LSTM models to learn meaningful temporal representations. Deep learning architectures have large numbers of trainable parameters and require substantial data to tune those parameters without overfitting. On a small dataset, the models tend to memorize noise rather than generalize from signal.

Second, and perhaps more critically, the data is not a single continuous time series. It is composed of four parallel city streams merged into one table, meaning that consecutive rows in the dataset often belong to different cities rather than representing a continuous temporal sequence for a single location. LSTM's core strength lies in its ability to carry information forward through time, but this advantage is entirely undermined when the temporal continuity of the input sequence is broken by city transitions at every fourth row.

Third, the CNN component of the CNN-LSTM model requires a sufficiently long input sequence to extract meaningful local patterns through its convolutional filters. With only seven timesteps per sample in the reshaped input tensor, the convolutional and pooling operations reduce the sequence to a length too short for the LSTM to process effectively, leading to the catastrophic performance reflected in the negative R^2 score.

4.2.4 Broader Implications

The results of this study reinforce a principle that is well-established in the machine learning literature but frequently overlooked in practice: **model complexity should be matched to data complexity.** A more sophisticated architecture does not guarantee better results. When the dataset is small, clean, and structurally simple, classical models trained with care will often outperform deep learning alternatives that require far more data to realize their potential.

This finding has direct practical relevance for researchers and practitioners working on environmental forecasting in data-scarce settings, where deploying an interpretable and accurate

Linear Regression model may be both more efficient and more reliable than pursuing deep learning solutions that the available data cannot adequately support.

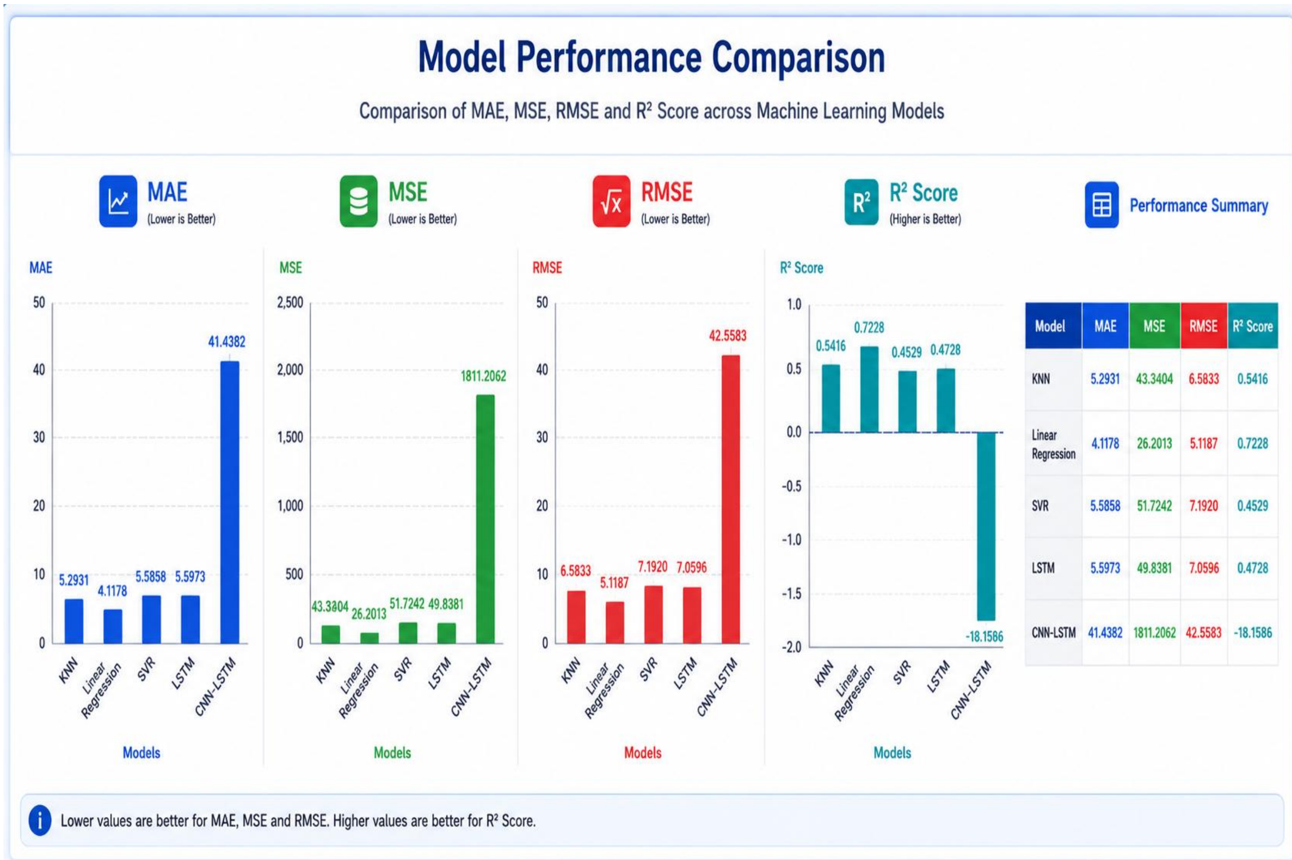

Figure 3 Model Performance Comparison

Chapter 5: Conclusion and Recommendation

5.1 Conclusion

This project set out to investigate the effectiveness of five machine learning regression models in predicting the Air Quality Index across four major Turkish cities, using daily environmental measurements of PM2.5, PM10, and temperature as the primary input variables. Through a structured pipeline of data preprocessing, feature engineering, model training, and comparative evaluation, the study produced results that are both practically meaningful and academically instructive.

The central finding of this work is that **Linear Regression emerged as the most effective model** for this particular dataset, achieving a testing R^2 of 72.28% and recording the lowest error values across all three error-based metrics. This outcome demonstrates that when the relationships between environmental variables and air quality are fundamentally linear in nature, and when the available dataset is moderate in size, a well-prepared classical model can deliver results that more complex architectures are unable to match.

K-Nearest Neighbors performed reasonably well as the second-best model with an R^2 of 54.16%, confirming that instance-based learning can capture local data patterns effectively in structured environmental datasets. Support Vector Regression, while theoretically equipped to model non-linear dynamics, achieved an R^2 of 45.29%, suggesting that the non-linear complexity it searches for was not sufficiently present in this dataset to justify its additional overhead.

The deep learning models — LSTM and CNN-LSTM — produced the weakest results of the five approaches evaluated. The CNN-LSTM model in particular recorded a deeply negative R^2 score, reflecting a fundamental mismatch between the architectural requirements of sequence-based deep learning and the structural characteristics of the dataset. These models demand large volumes of continuously ordered sequential data to function as intended, conditions that the current dataset, composed of merged daily records from four separate cities, could not fulfill.

Beyond the individual model results, this project demonstrates the full lifecycle of a data-driven air quality prediction system — from raw data ingestion and exploratory analysis through feature engineering, model training, evaluation, and live prediction. The live prediction interface developed as part of this work allows any user to input environmental readings and receive instant AQI estimates from all five models simultaneously, along with the corresponding EPA health category, making the system practically applicable beyond the academic context.

Taken together, the findings of this study affirm that machine learning holds genuine promise as a tool for urban air quality forecasting, while also highlighting that the selection of an appropriate model must be guided by a careful understanding of the data's size, structure, and underlying relationships rather than by architectural sophistication alone.

5.2 Recommendation

Based on the results and observations gathered throughout this project, the following recommendations are offered for researchers, practitioners, and institutions seeking to build upon this work:

1. Expand the dataset in both volume and feature richness. The most impactful improvement that could be made to this system is the incorporation of a larger and more feature-rich dataset. Adding environmental variables such as relative humidity, wind speed, nitrogen dioxide (NO_2), sulfur dioxide (SO_2), and carbon monoxide (CO) concentrations would provide the models — particularly SVR and the deep learning architectures — with a richer set of signals to learn from, likely improving prediction accuracy across all five approaches.

2. Restructure data for proper sequential modeling. For LSTM and CNN-LSTM models to perform as intended, the dataset should be reorganized so that each city is treated as an independent time series rather than being merged into a single mixed table. Training separate sequential models per city, or using a multi-variate time series framework with city as a conditioning variable, would allow these architectures to leverage their temporal memory capabilities properly and produce more meaningful results.

3. Perform systematic hyperparameter optimization. The models in this study were trained using reasonable default configurations. Future work should incorporate structured hyperparameter tuning — through grid search, random search, or Bayesian optimization — particularly for SVR's regularization parameter C and epsilon value, and for the learning rate, batch size, and number of units in the deep learning models. Even modest tuning improvements could meaningfully close the performance gap between models.

4. Extend geographic coverage. The current study is limited to four Turkish cities, which restricts the generalizability of its findings. Replicating this comparative analysis on datasets from cities in Africa, Asia, or other regions with distinct climatic and industrial profiles would help determine whether the dominance of Linear Regression observed here is specific to this dataset or reflects a broader pattern in structured urban air quality data.

5. Explore ensemble and hybrid classical approaches. Given the strong baseline performance of Linear Regression and the reasonable results from KNN, future work could investigate whether ensemble methods such as Random Forest Regression or Gradient Boosting offer further improvements over individual classical models. These approaches combine the interpretability of tree-based reasoning with greater flexibility than linear models, and tend to perform strongly on structured tabular datasets of the size used in this study.

6. Deploy the model as a real-time web application. The live prediction cell developed in this notebook represents a functional proof of concept. A natural next step would be to package the trained models into a web-based application — using a framework such as Flask, FastAPI, or Streamlit — that ingests real-time sensor data, generates continuous AQI forecasts, and delivers public-facing air quality alerts. Such a system would translate the academic findings of this project into a tangible tool for environmental monitoring and public health communication.

Figure 4 Smart Cities, Clean Air, Sustainable Future.

Final Remarks

This project contributes to the growing body of evidence that data-driven approaches to environmental monitoring are not only technically feasible but practically valuable. As urban populations continue to grow and the consequences of poor air quality become increasingly visible, the ability to forecast pollution levels accurately and in advance will be an essential component of any serious strategy for sustainable urban living. The work presented here, modest in scope but rigorous in method, offers a meaningful step in that direction.

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